INCH POUND

MIL-PRF-50204D (AR)
23 September 1997
SUPERSEDING
MIL-S-50204C (AR)
21 December 1993

PERFORMANCE SPECIFICATION

SUPPORTS, STRUCTURAL

(FOR USE IN AMMUNITION CONTAINERS)

This specification is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for structural supports for use inside ammunition containers.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are needed to meet the requirements specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army TACOM-ARDEC, ATTN: AMSTA-AR-EDE-S, Picatinny Arsenal, NJ 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8135
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:16Z
Check the source to verify that this is the current version before use.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

(Unless otherwise indicated, copies of the above specifications, standards and handbooks are available from DODSSP, Customer Service, Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5049.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B117 - Salt Spray (Fog) Testing, Method of

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Bar Harbor Drive, West Conshohocken, PA 19428-2959).

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:16Z
Check the source to verify that this is the current version before use.

3.2 Materials. Any material(s) that meets the requirements specified herein may be used.

3.3 Product acceptance. Quality plans and procedures used to assess conformance to the contract and product specifications shall be in accordance with MIL-STD-1916.

3.4 Operating requirements. Each support shall provide the following functional, operational, and performance capabilities.

3.4.1 Color. Unless otherwise specified, the finished support shall be of a dark coloration. Black, gray, brown, or dark green are acceptable. The support may also be of natural kraft paper coloration if constructed of kraft paper.

3.5 Interface requirements. The support shall accommodate the following interfaces.

3.5.1 Physical characteristics. The finished support must be dimensionally stable to interface with the ammunition container and its contents and not affect form, fit, and function of the container or item(s). The dimensions and tolerances of the finished part will be in accordance with the interface control drawing(s) or detailed design drawings specified in the contract.

3.6 Ownership and support requirements. Each support shall possess the following life-cycle ownership characteristics.

3.6.1 Safety. The support shall be physically safe to operate and handle with no additional safety concerns beyond that associated with item, container, and supports during storage, transport, and handling.

3.6.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the materials meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.7 Operating environment requirements. Each support shall be operable under the following environmental conditions without damage or loss in performance.

3.7.1 Water absorption. The finished component shall have a maximum water absorption of 10 percent by weight and shall show no evidence of cracking, splitting, warping, or breaking after being submerged under water for 48 hours minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:16Z
Check the source to verify that this is the current version before use.

3.7.2 Thermal stress cracking resistance. The finished part shall not crack, split, warp, break, become tacky or ooze or otherwise degrade when exposed to 160 degrees Fahrenheit for a duration of 100 hours minimum. The parts shall meet all dimensional requirements when returned to ambient conditions after exposure to high temperatures.

3.7.3 Rough handling performance.

3.7.3.1 Impact resistance (3 Foot). The structural component shall not crack, split, or break when subject to 3 foot drop testing.

3.7.3.2 Repetitive shock resistance. The structural component shall not crack, split, or break when subject to loose cargo testing.

3.7.3.3 Impact resistance (7 Foot). The structural component shall not crack, split, or break when subject to 7 foot drop testing.

3.7.4 Salt spray. The finished component shall be subjected to a 5 percent solution for 96 hours minimum and shall not corrode or affect form, fit and function.

3.7.5 Fungus. The support shall not exhibit excessive fungi growth after being subject to conditions favorable for fungi development, namely high humidity, warm atmosphere, and presence of inorganic salts.

4. VERIFICATION

4.1 General provisions. The inspections (examinations and tests) herein shall be performed to determine whether or not the item conforms to section 3 of this specification.

4.1.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)

b. Conformance inspection (see 4.3).

4.1.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified herein.

4.1.3 Classification of characteristics. For examinations and tests cited herein or when required by contract; critical, major, and minor characteristics are as defined in Section 3 of MIL-STD-1916.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:16Z
Check the source to verify that this is the current version before use.

TABLE I. Requirement/verification matrix

VERIFICATION METHODS

N/A - NOT AVAILABLE
1 - ANALYSIS
2 - DEMONSTRATION
3 - EXAMINATION
4 - TEST

VERIFICATION CLASSES

A = FIRST ARTICLE
B = CONFORMANCE

VERIFICATION LEVEL (VL)

SECTION 3 REQUIREMENT	VERIFICATION METHOD					VERIFICATION CLASS		SECTION 4 VERIFICATION	VL
	N/A	1	2	3	4	A	B		
3.4		X				X		4.4	
3.4.1				X		X	X	4.4.1	III
3.5		X				X		4.5	
3.5.1				X		X	X	4.5.1	IV
3.6		X				X		4.6	
3.6.1				X		X	X	4.6.1	III
3.7		X				X		4.7	
3.7.1					X	X		4.7.1	
3.7.2					X	X		4.7.2	
3.7.3					X	X		4.7.3	
3.7.3.1					X	X		4.7.3.1	
3.7.3.2					X	X		4.7.3.2	
3.7.3.3					X	X		4.7.3.3	
3.7.4					X	X		4.7.4	
3.7.5					X	X		4.7.5	

4.2 First article inspection.

4.2.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article samples shall consist of a minimum of 12 sample structural supports to perform the specified examinations and tests when required by contract (see 6.2).

4.2.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified (see Table I). Unless otherwise specified, all the inspections shall be performed.

4.2.3 Rejection. If any test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of a test specimen to comply with any of the requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:16Z
Check the source to verify that this is the current version before use.

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Lot formation shall be in accordance with Section 4 of MIL-STD-1916.

4.3.2 Classification of characteristics. Conformance examinations and tests are specified in Table I. When cited herein, attribute sampling inspection shall be conducted in accordance with MIL-STD-1916 using the verification levels stated in Table I unless otherwise specified in the contract or purchase order.

4.3.3 Testing.

4.3.3.1 Conditioning. All configurations (container, item, structural component(s) and supporting dunnage) shall be conditioned to a temperature of -65 (plus or minus 5 degrees) Fahrenheit or +160 (plus or minus 5 degrees) Fahrenheit for 24 hours prior to any rough handling testing. Each series of tests must begin within 15 minutes after removal of the configuration from the conditioning chamber. All tests must be completed within 35 minutes of start.

4.3.3.2 Methods of inspection (see 6.3).

4.4 Operating requirements verification. Complete each verification under 4.4.

4.4.1 Color verification. Unless otherwise specified, the finished support shall be visibly inspected to insure a dark coloration. Black, gray, brown, or dark green are acceptable. If the support is constructed of kraft paper products, the natural "kraft" color is also acceptable and will be visibly inspected for.

4.5 Interface requirements verification. Complete each verification under 4.5.

4.5.1 Physical characteristics. The finished support shall be examined for compliance with the applicable interface control drawings (see 6.2) using appropriate inspection equipment.

4.6 Ownership and support requirements. Complete each verification under 4.6.

4.6.1 Safety. The finished support shall be visually inspected to ensure they are free of hazardous burrs, nicks, sharp edges, foreign materials, or other imperfections posing physical danger to operators.

4.7 Operating Environment Requirements. Complete each verification under 4.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:16Z
Check the source to verify that this is the current version before use.

4.7.1 Water absorption. Under ambient atmospheric conditions, weigh 6 individual finished components. Immediately thereafter immerse the components in clear tap water (pH 6.5 to 7.5) at room temperature for 48 hours minimum. If necessary, weight down the components to ensure total submersion. After removing the components from the water, immediately surface dry the specimens and reweigh them. Compute the moisture gain as a percentage of pre-immersion weight.

4.7.2 Thermal stress cracking resistance. Prior to testing, visually inspect 6 finished components for any anomalies such as cracking, pitting etc. that have already passed dimensional examination. Subject the components to 100 hours minimum at 160 plus or minus 5 degrees Fahrenheit. Immediately following the time period, while still warm, visually examine each component for evidence of cracks, split, warping or other deformations. When the components return to ambient conditions perform dimensional check and insure the component is within tolerance.

4.7.3 Rough handling performance. Test samples for testing defined in 4.7.3.1, 4.7.3.2 and 4.7.3.3 will be conditioned in accordance with 4.3.3.1. The testing will be performed sequentially and at the two temperature extremes. Each temperature sequence will require 6 complete configurations (container, item(s), structural support(s), and associated dunnage, see 6.2). The first test will be a series of 3 foot drop testing in accordance with 4.7.3.1. Upon conclusion of the test, the container will be accessed and the structural component examined for performance as defined in the applicable requirement. The containers will then be re-packed and conditioned in accordance with 4.3.3.1. After proper conditioning at the appropriate temperature extreme, loose cargo testing will be performed in accordance with 4.7.3.2. Upon conclusion of the test, the containers will be accessed and the structural components examined for performance as defined in the applicable requirement. The containers will then be re-packed and conditioned in accordance with 4.3.3.1. After proper conditioning at the appropriate temperature extreme, 7 foot drop testing will be performed in accordance with 4.7.3.3. Upon completion of the test, the containers will be accessed and the structural components examined for performance as defined in the applicable requirement.

4.7.3.1 Impact resistance (3 foot). The orientations of the drops will be consistent with Figure 1. Each fully loaded container will be dropped in all the appropriate orientations depending on whether the configuration is cylindrical or rectangular. The container will be suspended in such a way that the lowest portion of the container (the desired impact area) is 3 feet from the impact surface. The container will then be released and allowed to fall uninhibited to the impact surface. The impact surface will be of a non-yielding substance similar to concrete or steel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:16Z
Check the source to verify that this is the current version before use.

4.7.3.2 Repetitive shock resistance (loose cargo). The loaded container shall be placed on the steel mounting surface (table) of the vibration equipment. The container shall not be restrained during vibration except by a fence attached to the test surface to prevent the package from falling off the table. A total free space between the container and the opposing side end boards will not exceed 2 inches. Rectangular containers shall be tested for 15 minutes on each of the most vulnerable horizontal and vertical faces on the same package (two faces total). Cylindrical containers shall be tested for 15 minutes on the bottom face (or top, if more vulnerable) and for 15 minutes on the most vulnerable circumferential position. If the container shifts circumferentially during the test, it shall be allowed to do so. The vibratory frequency shall be 5 Hz and the vibratory surface shall have a 1 inch double amplitude. Total test time shall be 30 minutes for each container. Depending on the container size relative to the mounting surface, several containers may be tested simultaneously provided the fencing does not exceed 2 inches on both sides when the containers are positioned together.

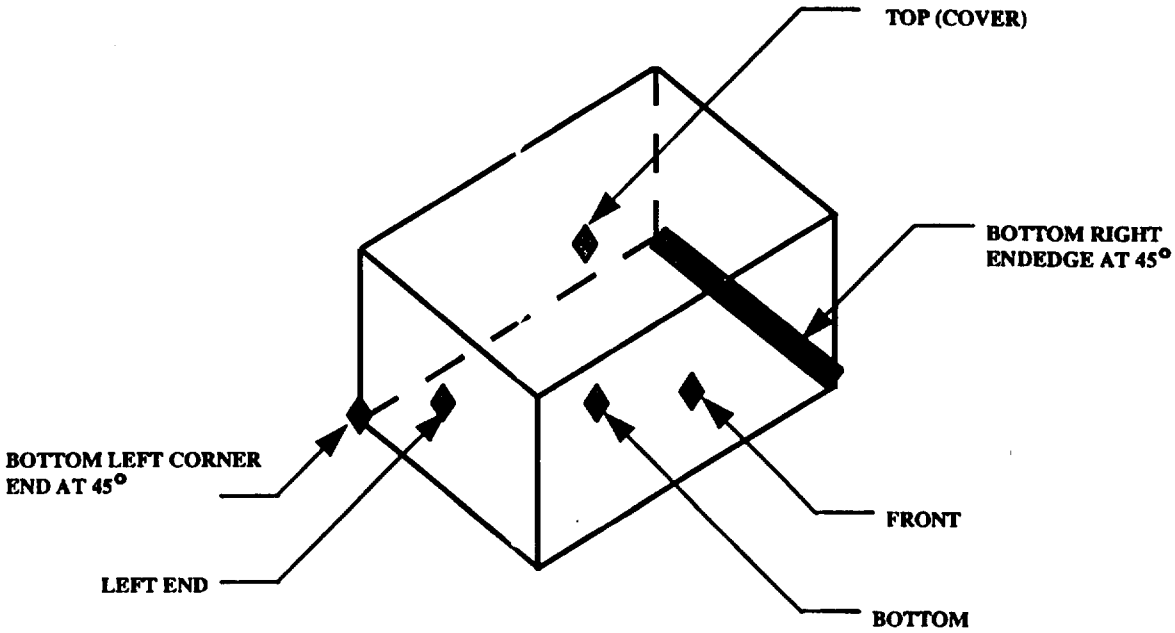
4.7.3.3. Impact resistance (7 foot). The orientations of the drops will be consistent with Figure 1. Each fully loaded container will be dropped only once, one container in each of the six orientations. The container will be suspended in such a way that the lowest portion of the container (the desired impact area) is 7 feet from the impact surface. The container will then be released and allowed to fall uninhibited to the impact surface. The impact surface will be of a non-yielding substance similar to concrete or steel.

4.7.4 Corrosion resistance. The components shall be salt spray tested in accordance with ASTM-B117.

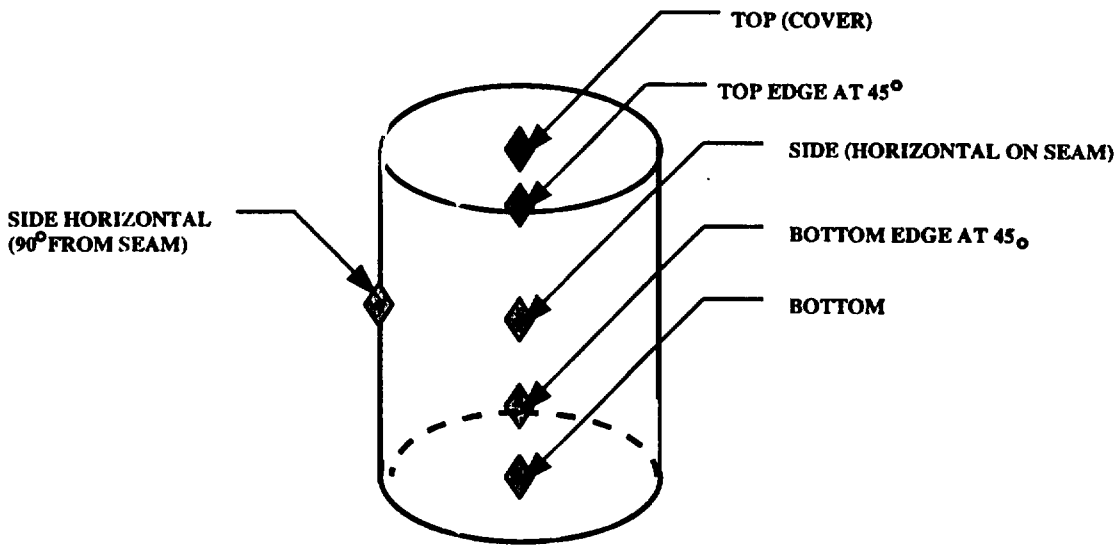
4.7.5 Fungus. The support shall be tested in accordance with Method 508 of MIL-STD-810.

4.7.6 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.3.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See 6.4 herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:16Z
Check the source to verify that this is the current version before use.



RECTANGULAR CONTAINER



CIRCULAR CONTAINER

FIGURE 1. Orientations for 3 & 7 Drop Testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:16Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the contract or order shall specify packaging requirements (see 6.2). When DOD personnel perform material packaging, those personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. The Inventory Control Point packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command, maintains packaging requirements. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

6.1 Intended use. The supports covered by this specification are intended to support and protect various types of munitions from damage due to shock and vibration when packaged in ammunition containers.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required the specific issue of individual documents referenced (see 2.3).

c. Requirements for submission of first article sample.

d. Container and item dimensions, tolerances, and interface control drawings.

e. Acquisition of inert ammunition items, containers marking requirements if necessary, and drawings to support rough handling tests.

6.3 Equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, U.S. Army TACOM-ARDEC, ATTN: AMSTA-AR-QAT-P, Picatinny Arsenal, NJ 07806-5000. This description should include, but not be limited to the accuracy and precision of the method, test data demonstrating the accuracy and precision, and drawings of any special equipment required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:16Z
Check the source to verify that this is the current version before use.

6.4 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army TACOM-ARDEC, ATTN: AMSTA-AR-QAT-P, Picatinny Arsenal, NJ 07806-5000. The address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.5 Subject term (key word) listing.

Packaging ammo
Projectiles

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
ARMY - AR

Preparing activity:
ARMY - AR

(Project 8135-0711)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:16Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced documents(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-PRF-50204D (AR)	970923

3. DOCUMENT TITLE

SUPPORTS, STRUCTURAL (FOR USE IN AMMUNITION CONTAINERS)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)*	
	(1) Commercial	(2) AUTOVON
U. S. Army ARDEC	(201) 724-6671	880-6671
c. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:	
ATTN: AMSTA-AR-EDE-S, B-12 Picatinny Arsenal, NJ 07806-5000	Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:16Z
Check the source to verify that this is the current version before use.